Transverse divergences of the terms in your `Other_terms.tex`

Your file without Group I Row II part 1 and Group II Row IV (both inside `\begin{comment}`). Question: does any term diverge when transverse coordinates come together (UV), or anywhere else in the transverse plane? $\mathcal{K}(X) = X/X^2$, $a = \frac{k^+}{\vee}$, $U^{bc} = U^{cb}$.

1 Answer

Flagged.

F1 Group I Row V, $d^3N_{2b\,\mathrm{UV}}$: UV divergent. The $\frac{2}{\epsilon}$ is the dimensionally regularized UV divergence of 2b at $x \to z$ (the two gluons of the loop at one point). After it, $\log \frac{(y-z)^2\mu^2}{4}$ times $\frac{(y-z)^i}{(y-z)^2}[U(y) - U(z)]$ is integrable at $y \to z$. In this file its partner is gone: the $\frac{11}{3}$ term of Group I Row II part 1, which cancels its β_0 pole, is commented out.

F2 Group I Row III: each of the parts 2, 3a, 3b is divergent; only the sum of the four parts is finite.

- At $z \to w$, each of parts 2, 3a, 3b grows like r^{-2} (logarithmic divergence).
- Across the curve $(x-z)^2 = (x-w)^2$ (it passes through $z = w$), the separate lines go like $[(x-z)^2 - (x-w)^2]^{-2}$, which is not integrable.
- Across the segment where w lies between x and z, the $\mathcal{A}_0$ lines go like 1/distance, which is not integrable either.
- The sum of parts 1, 2, 3a, 3b is regular on both curves, and its worst UV behaviour is r^{-1} (at $x \to w$), which is integrable.

Row III must be kept as one expression.

F3 Group I Row V, $d^3N_{2b\,\mathrm{finite}}$: separate lines divergent at $x \to z$, the sum finite. The $d_\perp$ line, the line without the plus prescription, and the $\frac{11}{6}$ line each go like $(x-z)^{-2}$. The $\frac{11}{6}$ line is the subtraction that removes this divergence.

F4 Group II Row I, d^3N_{2f}: each of its three lines is divergent, the sum finite. Line 2, $\left[\frac{1}{\vee} - \frac{1}{k^+}\right]\frac{1}{(z-w)^2(z-w')^2}$, diverges logarithmically at $z \to w$ and at $z \to w'$. Line 1 diverges at $z \to w$ and line 3 at $z \to w'$, and they cancel it. Keep the three lines together.

F5 Group I Row II, d^3N_{2b} (momentum form, R^i): UV finite, but infrared divergent (not UV). Two regions are affected.

- At $p \to 0$: $R^i \to -2k^i \log \frac{p^2}{k^2}$.
- At $q = k - p \to 0$: $R^i \to \frac{2k^2k^i}{q^2} \log \frac{k^2}{q^2}$.

With the $\frac{1}{p^2k^2}$, $\int d^2p$ diverges like $\log^2$ at both points. These are the regions $|x-y| \to \infty$. In addition, $\int d^2k$ uses the same letter k as the measured gluon.

Not divergent (UV and on curves): every other equation, i.e. Group I Rows III (sum) and IV, Row V first part and d^3N_{2a}, Group II Row I (every equation; separate lines of the second part need a principal value across the curve $(x'-w')^2(x-z)^2 = (x-w)^2(x'-z)^2$), Group III Rows I, II, III, V, VI.

2 Method

- Your equations are evaluated pointwise as written (each equation split into its lines, `terms.py`). Wilson lines and charges are smooth functions of position:
  - $U(p)$ is the adjoint of $V(p) = W(p)\,\mathrm{diag}(1,-1,-1)\,W(p)^\dagger$, with $W(p) = e^{i\theta_a(p)t^a}$; this is an SU(3) matrix and $U = U^T$.
  - $\theta_a(p)$ and $\rho^a(p)$ are sums of sines with wave numbers of order 1.

A colour factor that vanishes when two points coincide therefore vanishes linearly, as it does in the real integral.

- **UV.** For every pair of points P, Q, set $P = Q + r\,\hat{n}(\varphi)$ and average the integrand over φ, giving $A(r)$ at $r = 10^{-1.5} \dots 10^{-4.5}$. The exponent s in $A \sim r^s$ decides: $\int d^2r$ diverges iff $s \le -2$.
  - Where a singular curve crosses the circle, the φ average is a principal value.
  - Every equation was scanned, and every row with several equations as a whole.
  - Two field and point configurations were used. Terms with p^+ or q^+ integrals are evaluated at fixed p^+ ($0.37k^+$; the plus prescriptions subtract at $p^+ = 0$), on the pairs of points that can diverge (Sec. 5).
  - In Group I Row V $d^3N_{2b\,\text{finite}}$ the lines have different longitudinal variables (p^+, q^+, none), so they can cancel only after the integrals. There the p^+ and q^+ integrals are done (16-point Gauss–Legendre on $[0, k^+]$, with the orientation and $\frac{1}{2\pi}$ of each), `intcheck.py`.
- **Singular curves.** A point is moved across the curve; from $g(\pm\delta)$ take $E = \frac{g_+ + g_-}{2} \sim \delta^{e_E}$ and $O = \frac{g_+ - g_-}{2} \sim \delta^{e_O}$.
  - Integrable: $e_E > -1$ and $e_O > -1$.
  - Principal value: $e_O = -1$.
  - Divergent: $e_E \le -1$ or $e_O \le -2$.

3 UV: coincident points

s = exponent of $A(r)$ at the worst pair (all configurations and p^+ values); $s \le -2$ = divergent.

equation	worst s	pair	divergent pairs	verdict	scan
Group I, Row III, eq. 1	-0.12	$x \to w$	—	finite	all pairs, 2 configurations
Group I, Row III, eq. 2	-2.07	$z \to w$	$z \to w$	divergent	all pairs, 2 configurations
Group I, Row III, eq. 3	-2.07	$z \to w$	$z \to w$	divergent	all pairs, 2 configurations
Group I, Row III, eq. 4	-2.00	$z \to w$	$z \to w$	divergent	all pairs, 2 configurations
Group I, Row III **row total**	-1.01	$x \to w$	—	finite	all pairs, 2 configurations
Group I, Row IV, eq. 1	0.00	$x \to y$	—	finite	all pairs, 2 configurations
Group I, Row V, 1st part, eq. 1	-0.00	$x \to y$	—	finite	selected pairs, $p^+ = 0.37k^+$, line by line
Group I, Row V, 2nd part, eq. 1	-0.00	$x \to y$	—	finite	selected pairs, $p^+ = 0.37k^+$, line by line
Group I, Row V, 2nd part, eq. 3	-0.00	$x \to z$	—	finite	p^+, q^+ integrals done; lines 1, 3, 4 diverge at $x \to z$
Group II, Row I, 1st part, eq. 1	-0.00	$x \to y$	—	finite	all pairs, 2 configurations
Group II, Row I, 1st part, eq. 2	-0.00	$x \to y$	—	finite	all pairs, 2 configurations
Group II, Row I, 2nd part, eq. 1	-0.12	$x \to w$	—	finite	all pairs, 2 configurations
Group II, Row I, 2nd part, eq. 2	-0.13	$x' \to w'$	—	finite	all pairs, 2 configurations
Group II, Row I, 2nd part, eq. 3	-0.00	$x \to z$	—	finite	all pairs, 2 configurations
Group II, Row I, 2nd part, eq. 4	-0.13	$x' \to w'$	—	finite	all pairs, 2 configurations
Group II, Row I, 2nd part, eq. 5	-0.00	$x \to z$	—	finite	all pairs, 2 configurations
Group II, Row I, 2nd part, eq. 6	-0.01	$z \to w'$	—	finite	all pairs, 2 configurations; line 1: $z \to w$; line 2: $z \to w$, $z \to w'$; line 3: $z \to w'$
Group II, Row I **row total**	-0.00	$x \to y$	—	finite	all pairs, 2 configurations
Group III, Row I, eq. 1	0.00	$x \to y$	—	finite	all pairs, 2 configurations
Group III, Row II, eq. 1	0.00	$x \to y$	—	finite	all pairs, 2 configurations
Group III, Row III, 1st part, eq. 1	-0.00	$x \to z$	—	finite	selected pairs, $p^+ = 0.37k^+$, line by line
Group III, Row III, 2nd part, eq. 1	0.00	$x \to z$	—	finite	selected pairs, $p^+ = 0.37k^+$, line by line
Group III, Row V, eq. 1	-0.00	$x \to z$	—	finite	selected pairs, $p^+ = 0.37k^+$, line by line
Group III, Row VI, eq. 1	-0.00	$x \to z$	—	finite	selected pairs, $p^+ = 0.37k^+$, line by line

Exponents near 0 or -1 are bounded or $1/r$ singularities, both integrable. Group I Row II d^3N_{2b} (momentum integrals) and Row V $d^3N_{2b\,\text{UV}}$ (explicit pole) are not scanned; see F5 and F1.

4 Singular curves

equation	curve	separate lines	sum: e_E, e_O	sum
Group I, Row III, eq. 1	$(x-z)^2=(x-w)^2$	DIVERGENT, integrable	$e_E = 0.00$, $e_O = -1.00$	principal value
Group I, Row III, eq. 2	$(x-z)^2=(x-w)^2$	DIVERGENT, principal value	$e_E = -2.11$, $e_O = -1.17$	DIVERGENT
Group I, Row III, eq. 3	$(x-z)^2=(x-w)^2$	DIVERGENT, principal value	$e_E = -2.11$, $e_O = -1.10$	DIVERGENT
Group I, Row III, eq. 3	w on the segment $[x,z]$ ($\mathcal{A}_0$)	DIVERGENT	$e_E = -1.00$, $e_O = -0.01$	DIVERGENT
Group I, Row III, eq. 4	w on the segment $[x,z]$ ($\mathcal{A}_0$)	DIVERGENT	$e_E = -1.00$, $e_O = -0.00$	DIVERGENT
Group II, Row I, 2nd part, eq. 1	$(x-z)^2=(x-w)^2$	integrable	$e_E = -0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 1	$(x'-w')^2(x-z)^2=(x-w)^2(x'-z)^2$	integrable	$e_E = 0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 2	$(x'-w')^2(x-z)^2=(x-w)^2(x'-z)^2$	integrable, principal value	$e_E = -0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 3	$(x-z)^2=(x-w)^2$	integrable	$e_E = 0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 3	$(x'-w')^2(x-z)^2=(x-w)^2(x'-z)^2$	principal value	$e_E = -0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 4	$(x-z)^2=(x-w)^2$	integrable	$e_E = 0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 4	$(x'-w')^2(x-z)^2=(x-w)^2(x'-z)^2$	principal value	$e_E = -0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 4	$(x'-w')^2=(x'-z)^2$	integrable	$e_E = -0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 5	$(x-z)^2=(x-w)^2$	integrable	$e_E = -0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 5	$(x'-w')^2(x-z)^2=(x-w)^2(x'-z)^2$	principal value	$e_E = 0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 5	$(x'-w')^2=(x'-z)^2$	integrable	$e_E = -0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 6	$(x-z)^2=(x-w)^2$	integrable	$e_E = 2.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 6	$(x'-w')^2(x-z)^2=(x-w)^2(x'-z)^2$	principal value	$e_E = -0.00$, $e_O = 1.00$	integrable
Group II, Row I, 2nd part, eq. 6	$(x'-w')^2=(x'-z)^2$	integrable	$e_E = 1.80$, $e_O = 1.00$	integrable
Group III, Row III, 2nd part, eq. 1	$(x-z)^2=(x-w)^2$	integrable	$e_E = 0.00$, $e_O = 1.00$	integrable
Group I, Row III **row total**	$(x-z)^2=(x-w)^2$	DIVERGENT, integrable, principal value	$e_E = 0.00$, $e_O = 0.99$	integrable
Group I, Row III **row total**	w on the segment $[x,z]$ ($\mathcal{A}_0$)	DIVERGENT	$e_E = -0.00$, $e_O = -0.00$	integrable
Group II, Row I **row total**	$(x-z)^2=(x-w)^2$	integrable	$e_E = 0.00$, $e_O = 1.00$	integrable
Group II, Row I **row total**	$(x'-w')^2(x-z)^2=(x-w)^2(x'-z)^2$	integrable, principal value	$e_E = -0.00$, $e_O = 1.00$	integrable
Group II, Row I **row total**	$(x'-w')^2=(x'-z)^2$	integrable	$e_E = -0.00$, $e_O = 1.00$	integrable
Group III, Row III **row total**	$(x-z)^2=(x-w)^2$	integrable	$e_E = 0.00$, $e_O = 1.00$	integrable

5 Analytic checks

Power counting. In two dimensions $\int d^2r\, r^{-n}$ converges at $r \to 0$ for $n < 2$. When m points collapse together the measure is $r^{2m-3}dr$. Each $\mathcal{K}(u-v)$ contributes r^{-1} when $u \to v$. A colour bracket like $U(z)-U(y)$ adds a factor r when $y \to z$.

- **Group III Rows III, V, VI, line 1** (p^+ integral) has $\mathcal{K}^m(y'-z)\,\mathcal{K}^{k'}(x'-y')\,\mathcal{K}^i(y-w)\,\mathcal{K}^j(x-z)$. Each kernel joins a different pair, so every two-point limit is r^{-1}. A collapse of m points brings at most $m-1$ kernels, i.e. $r^{-(m-1)}$ against r^{2m-3}. The same holds for line 2 ($\mathcal{K}(x'-z)$ in place of $\mathcal{K}(x'-y')$) and line 3. Hence these rows are finite.

- **Group III Row III, second part.** $p^+(x-z)^2+k^+(x-w)^2$ vanishes only if x, z, w all coincide. Its $\frac{1}{(z-w)^2}$ always multiplies $(z-w)$ or $(x-z)^2-(x-w)^2$, which is $O(r)$ at $z \to w$.

- **Group I Row V, d^3N_{2a}.** At $x \to z$ the $d_\perp$ line goes like $\frac{(z-x)^i}{(x-z)^2}\frac{(y-x)^2-(y-z)^2}{(x-z)^2} \sim r^{-2}$, with a non-zero angular

average. But its colour factor vanishes there: $f^{adc}U^{db}(z)U^{ce}(z) = f^{hbe}U^{ah}(z)$ and $f^{fbe}f^{hbe} = N_c\delta^{fh}$. So

$$\big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_cU^{af}(x)\big]_{x=z} = -N_cU^{af}(z) + N_cU^{af}(z) = 0,$$

and the integrand is $O(r^{-1})$. The same holds for the q^+ line.

- **Group I Row V, $d^3N_{2b\,\text{finite}}$.** Its colour factor $N_c[U(y) - U(x)]$ does not vanish at $x \to z$.
  - The $d_\perp$ line and the q^+ line without the plus prescription go like $(x-z)^{-2}$, with a non-zero average.
  - In the $[\frac{k^+}{q^+}]_+$ line the leading term does not depend on q^+ at $x = z$, so the plus prescription removes it.
  - The $\frac{11}{6}$ line, with $\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2} \to \frac{1}{(x-z)^2}$, cancels the rest. This is the subtraction of `rowV/Part2b_MSbar.pdf`: the sum is finite for every p^+.

Group I Row III, the $\mathcal{A}_0$ segment. $\mathcal{A}_0 = \int_0^1 \frac{dt}{|(x-w)-t(x-z)|^2}$, which is $\simeq \frac{\pi}{|(x-w)\times(x-z)|}$ when w lies on the segment from x to z: a 1/distance singularity across a line, not integrable. On the segment $x - w = t\,e$, $z - w = -(1-t)e$, with $e = x - z$. Let $Q \equiv \frac{[(x'-w')\cdot e][(y-z)\cdot e]}{e^2}$, and drop the common $\frac{1}{(x'-w')^2(y-z)^2}$. The four $\mathcal{A}_0$ lines (3 in part 3a, 1 in part 3b) become

$$\underbrace{-\frac{t}{1-t}Q - \frac{1}{1-t}Q + \frac{t}{1-t}Q}_{\text{3a}} \underbrace{+\frac{1}{1-t}Q}_{\text{3b}} = 0.$$

So their sum vanishes linearly on the segment and $\mathcal{A}_0\times$(sum) is bounded. Part 3a or part 3b alone is not: they must stay together.

Group I Row II, R^i. Limits, checked numerically to 10^{-6}:

- $p \to \infty$: $R^i \to k^i[4\log a - \log(1-a)]$, giving $\int \frac{d^2p}{p^2}e^{ip\cdot(y-x)} \sim -2\pi\log|x-y|$ (integrable).
- $k \to \infty$: $R^i \to 2k^i\log\frac{k^2}{p^2}$, giving $\frac{(w-y)^i}{(w-y)^2}\log|w-y|$ (integrable).
- $x, y \to w$ together: the transform is homogeneous of degree -1, against r^3dr (integrable).
- $p \to 0$ and $p \to k$: the $\log^2$ infrared divergences of F5.

Reproducing

In `nlo_gluon_paper/uv_check/`: `uvscan2.py` (all pairs, equations and rows), `uvtarget.py` (rows with p^+ integrals), `intcheck.py` (Row V $d^3N_{2b\,\text{finite}}$ with the integrals done), `surfscan.py` (singular curves), `make_report.py` (this file).